TELIPORT
Script Your Future
Season 4

TATA
TATA ELXSI

Case Study 2

Team StratWise

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THINK. BUILD. CREATE. WIN

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Season 4

Test a battery before you build it

VOLT-FORGE connects your battery design decisions to what they physically change

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PROBLEM UNDERSTANDING

Cells rarely share the load evenly

Say a Raspberry Pi 5 gateway must survive a **10-minute power cut at 25 W**, its full supply rating, at 23 °C. Four cells in parallel sounds easy, until the cell closest to the output takes more current at first and warms sooner. Ageing and packaging come later.

DESIRABILITY

You shouldn't need a lab to learn that

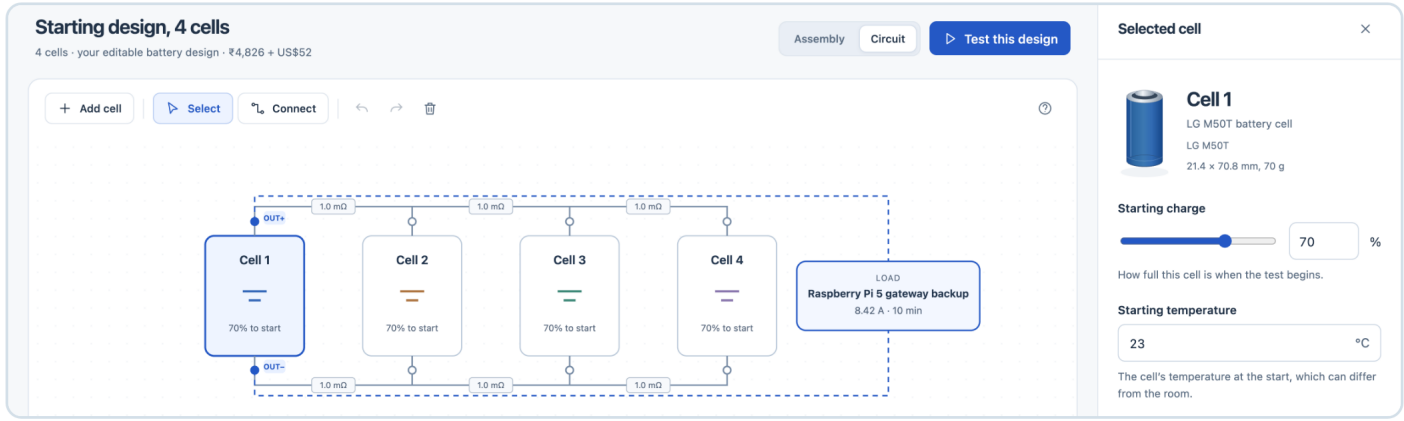
Today that takes a bench, a paid simulation tool or your own code. Students can't afford that and engineers can't wait. Here you wire the circuit on screen, run it and see which cell struggles **before spending a rupee**.

TECHNICAL FEASIBILITY

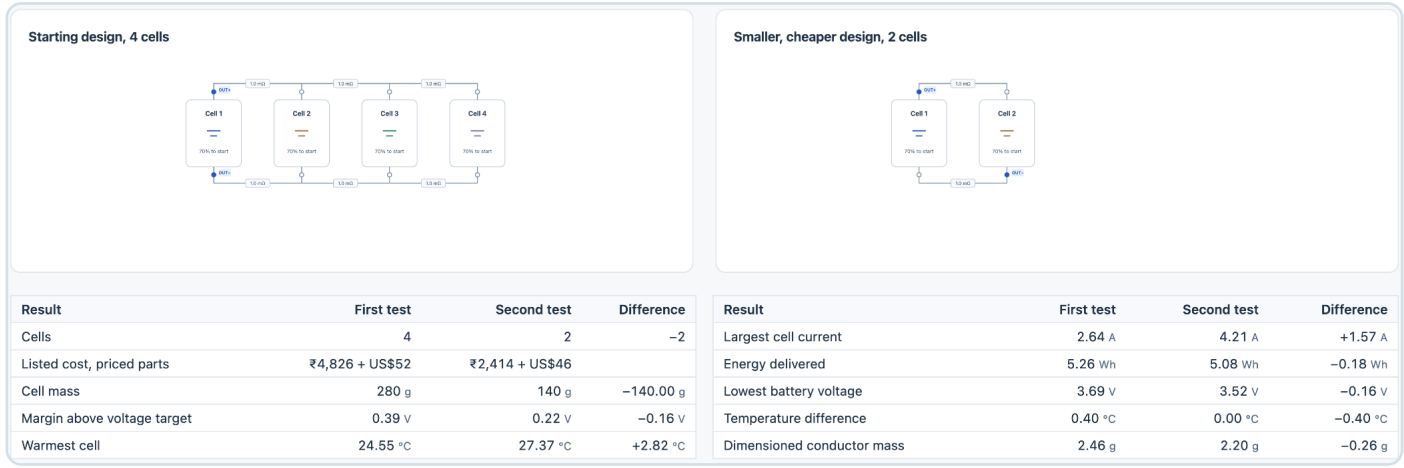
So how few cells can do this job?

We compared 3 and 2 cells against the 4-cell start. **Two cells** held the Pi for 10 minutes in all 33 simulated tests, even with one weak cell. One cell was refused. It should fit a two-cell Pi UPS board, though we haven't built it, and the tool runs on a laptop.

PROTOTYPE & EVIDENCE Test it out: **volt-forge-app.vercel.app**



Build it. This is the real circuit the solver runs. Click any cell to see and change what's inside it.



Result	First test	Second test	Difference
Cells	4	2	−2
Listed cost, priced parts	₹4,826 + US$52	₹2,414 + US$46	
Cell mass	280 g	140 g	−140.00 g
Margin above voltage target	0.39 V	0.22 V	−0.16 V
Warmest cell	24.55 °C	27.37 °C	+2.82 °C

Result	First test	Second test	Difference
Largest cell current	2.64 A	4.21 A	+1.57 A
Energy delivered	5.26 Wh	5.08 Wh	−0.18 Wh
Lowest battery voltage	3.69 V	3.52 V	−0.16 V
Temperature difference	0.40 °C	0.00 °C	−0.40 °C
Dimensioned conductor mass	2.46 g	2.20 g	−0.26 g

Compare it. Same 25 W job: 2 cells cost and weigh half, but work harder with less voltage margin.

4,006 simulated cases. 13 designs against weak cells, bad links and poor cooling. A full rerun matched exactly.

54 published four-cell experiments. All 194 dataset files verified intact, 2 tests compared with the model so far.

15 of 15 solver checks. PyBaMM, an independent solver, matched our voltage and heat.

Every cost, source and check is in the full appendix: **volt-forge-app.vercel.app/appendix**

ENGINEERING QUALITY

Why should you trust it?

Every run must balance its energy, which catches solver errors, not a wrong model. Against published data, voltage and temperature missed all 12 checks. Current sharing came closer but also missed: the nearest cell took **34%** early on, the model said **34%**. Over 10 minutes, real heating stayed within 1.13 times the model.

INNOVATION

You don't have to guess

Press **Find the smallest design**. It rebuilds the circuit at each size and tests it under normal conditions. For the Pi: 2 cells, not 4, at half the cell cost. Our study then stress-tested that.

BUSINESS VIABILITY



41.3 lakh TAM, engineering students

15.6 lakh SAM, electrical, electronics, mechanical

130 SOM, labs at ₹50,000, 5% assumed

₹35 crore

to build it all. Stage one is ₹11 crore: 130 labs and 31 teams at ₹5 lakh a year would repay it in 5 years, before running costs. Students use it free. No customer has seen these prices.

Simulated with an assumed 90% converter and 3.3 V floor. **Next:** a bench test with voltage, current and heat cutoffs, then student and engineer trials.